1.

P	**Q**	**¬P**	**¬Q**	**P ⇒ ¬Q**	**Q ⇒ ¬P**
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

Since the last two columns have the same values, P ⇒ ¬Q is equivalent to Q ⇒ ¬P.

P	**Q**	**¬P**	**¬Q**	**P ⇒ ¬Q**	**¬Q ⇒ P**	**P ⇔ ¬Q**	**P ∧ ¬Q**	**¬P ∧ Q**	**((P ∧ ¬Q) ∨ (¬P ∧ Q))**
T	T	F	F	F	T	F	F	F	F
T	F	F	T	T	T	T	T	F	T
F	T	T	F	T	T	T	F	T	T
F	F	T	T	T	F	F	F	F	F

Since the seventh and last column have the same values, P ⇔ ¬Q is equivalent to ((P ∧ ¬Q) ∨ (¬P ∧ Q)).

2.

Smoke	**Fire**	**¬Smoke**	**¬Fire**	**Smoke ⇒ Fire**	**¬Smoke ⇒ ¬Fire**	**(Smoke ⇒ Fire) ⇒ (¬Smoke ⇒ ¬Fire)**
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

Since we have trues and falses in the last column, (Smoke ⇒ Fire) ⇒ (¬Smoke ⇒ ¬Fire) is neither valid nor unsatisfiable.

Smoke	**Fire**	**Heat**	**Smoke ⇒ Fire**	**Smoke v Heat**	**(Smoke v Heat) ⇒ Fire**	**(Smoke ⇒ Fire) ⇒ ((Smoke v Heat) ⇒ Fire)**
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	F	T	F	T
T	F	F	F	T	F	T
F	F	F	T	F	T	T
F	F	T	T	T	F	F
F	T	F	T	F	T	T
F	T	T	T	T	T	T

The statement given, (Smoke ⇒ Fire) ⇒ ((Smoke v Heat) ⇒ Fire), is neither valid nor unsatisfiable.

Smoke	**Fire**	**Heat**	**Smoke ∧ Heat**	**(Smoke ∧ Heat) ⇒ Fire**	**Smoke ⇒ Fire**	**Heat ⇒ Fire**	**((Smoke ⇒ Fire) ∨ (Heat ⇒ Fire))**	**((Smoke ∧ Heat) ⇒ Fire) ⇔ ((Smoke ⇒ Fire) ∨ (Heat ⇒ Fire))**
T	T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T	T
T	F	T	T	F	F	F	F	T
T	F	F	F	T	F	T	T	T
F	F	F	F	T	T	T	T	T
F	F	T	F	T	T	F	T	T
F	T	F	F	T	T	T	T	T
F	T	T	F	T	T	T	T	T

The statement ((Smoke ∧ Heat) ⇒ Fire) ⇔ ((Smoke ⇒ Fire) ∨ (Heat ⇒ Fire)) is valid since it is always true.

3. a.

Y: Mythical
I: Immortal
M: Mammal
H: Horned
G: Magical

Knowledge Base:

- $Y \Rightarrow I$
- $\neg Y \Rightarrow (\neg I \wedge M)$
- $(I \vee M) \Rightarrow H$
- $H \Rightarrow G$

b.

Get rid of $\Rightarrow$:

- $\neg Y \vee I$
- $Y \vee (\neg I \wedge M)$
- $\neg(I \vee M) \vee H$
- $\neg H \vee G$

Push negation Inwards (deMorgan's law):

- $\neg Y \vee I$
- $Y \vee (\neg I \wedge M)$
- $(\neg I \wedge \neg M) \vee H$
- $\neg H \vee G$

Distribute:

- $\neg Y \vee I$
- $(Y \vee \neg I) \wedge (Y \vee M)$
- $(\neg I \vee H) \wedge (\neg M \vee H)$
- $\neg H \vee G$

Final CNF: $(\neg Y \vee I) \wedge (Y \vee \neg I) \wedge (Y \vee M) \wedge (\neg I \vee H) \wedge (\neg M \vee H) \wedge (\neg H \vee G)$

$(\neg \text{Mythical} \vee \text{Immortal}) \wedge (\text{Mythical} \vee \neg \text{Immortal}) \wedge (\text{Mythical} \vee \text{Mammal}) \wedge (\neg \text{Immortal} \vee \text{Horned}) \wedge (\neg \text{Mammal} \vee \text{Horned}) \wedge (\neg \text{Horned} \vee \text{Magical})$

c.
Mythical:

1. $\neg$Mythical v Immortal
2. Mythical v $\neg$Immortal
3. Mythical v Mammal
4. $\neg$Immortal v Horned
5. $\neg$Mammal v Horned

6. ¬Horned v Magical
7. ¬Mythical (negation of what we want to prove)
8. ¬Immortal by 2,7
9. Mammal by 3,7
10. Horned by 5,9
11. Magical by 6,10

We did not encounter a contradiction, so we cannot prove KB entails mythical.

Horned:

1. ¬Mythical v Immortal
2. Mythical v ¬Immortal
3. Mythical v Mammal
4. ¬Immortal v Horned
5. ¬Mammal v Horned
6. ¬Horned v Magical
7. ¬Horned (negation of what we want to prove)
8. ¬Immortal by 4,7
9. ¬Mammal by 5,7
10. ¬Mythical by 1,8
11. Mammal by 3,10
12. Empty clause by 9,11

Found a contradiction when we added ¬Horned, so proved KB entails Horned.

Magical:

1. ¬Mythical v Immortal
2. Mythical v ¬Immortal
3. Mythical v Mammal
4. ¬Immortal v Horned
5. ¬Mammal v Horned
6. ¬Horned v Magical
7. ¬Magical (negation of what we want to prove)
8. ¬Horned by 6,7
9. ¬Immortal by 4,8
10. ¬Mammal by 5,8
11. ¬Mythical by 1,9
12. Mythical by 3,10
13. Empty clause by 11,12

Found a contradiction when we added ¬Magical, so proved KB entails Magical.

4.

Figure 1:

(((¬A ∧ B) v (¬B ∧ A)) ∧ (C v (¬D ∧ ¬C))) v (((¬A ∧ ¬B) v A) ∧ ((C ∧ ¬D) v (D ∧ ¬C)))

Decomposable: Yes. For any conjuncts, there are different variables on each side of the and symbol. Therefore, this is decomposable. var(α) ∩ var(β) = ⌀ always holds.

Deterministic: No. Some disjuncts are not mutually exclusive. Let A = True, B = False, C = True, and D = False. After substituting, we get:
(((¬T ⋀ F) v (¬F ⋀ T)) ⋀ (T v (¬F ⋀ ¬T))) v (((¬T ⋀ ¬F) v T) ⋀ ((T ⋀ ¬F) v (F ⋀ ¬T))) =
(T) v (T)

Since both sides of the disjunction (v) are evaluated as True, it is not mutually exclusive so it is not deterministic.

Smooth: No. It is not smooth since not all disjuncts mention the same set of variables. For example, in C v (¬D ⋀ ¬C), the left side has C while the right side has C and D. Similarly, in ¬A ⋀ ¬B) v A, the left side mentions A and B while the right only has A. var(α) ≠ var(β) holds for all disjuncts.

Figure 2:
((¬A ⋀ B) v (¬A ⋀ B)) ⋀ ((C ⋀ D) v (¬D ⋀ ¬C)) v ((¬A ⋀ B) v (¬A ⋀ B)) ⋀ ((C ⋀ ¬D) v (D ⋀ ¬C))

Decomposable: Yes. For any conjuncts, there are different variables on each side of the and symbol. Therefore, this is decomposable. var(α) ∩ var(β) = ⌀ always holds.

Deterministic: No. Some disjuncts are not mutually exclusive. We see the following disjunct twice: (¬A ⋀ B) v (¬A ⋀ B). If we let A = False and B = True, both sides would evaluate to true, so it is not deterministic since they are not mutually exclusive.

Smoothness: Yes. For all disjuncts, we see the same variables on both the left and right side. At the top-most or, we see A, B, C and D on each side. In the first half of the left side, we see a disjunct with A and B on both sides and after that on the left side we see a disjunct with C and D on each side. On the right-hand side overall, we see a disjunct with A and B on both sides and one with C and D on both sides. var(α) = var(β) holds for all disjuncts.

5.
a.

A	B	¬A	¬B	¬A ⋀ B	¬B ⋀ A	(¬A ⋀ B) ⋁ (¬B ⋀ A)
T	T	F	F	F	F	F
T	F	F	T	F	T	T
F	T	T	F	T	F	T
F	F	T	T	F	F	F

2nd and 3rd worlds are true
w(A)*w(¬B) + w(B)*w(¬A) = 0.1*0.7 + 0.3*0.9 = 0.34

b. We get the same answer, as shown below.

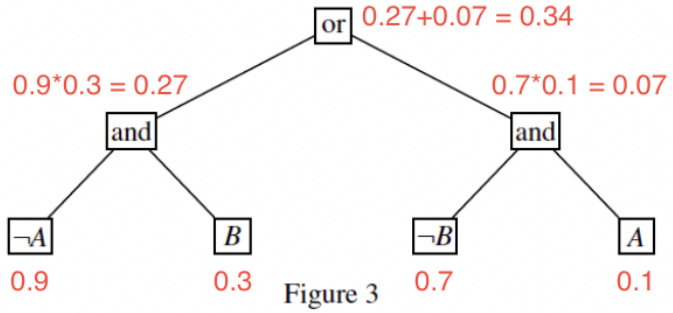
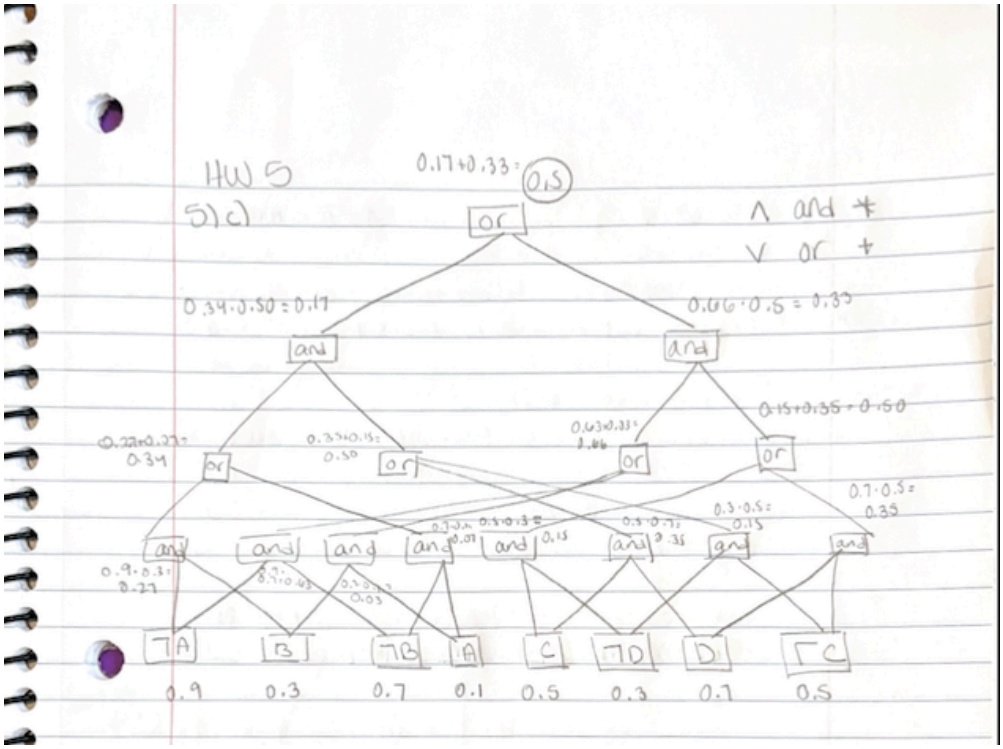


Figure 3

c. **0.5**



((w(¬A)*w(B)) + (w(¬B)*w(A))) * ((w(C)*w(D)) + (w(¬D)*w(¬C))) +
((w(¬A)*w(¬B)) + (w(B)*w(A))) * ((w(C)*w(¬D)) + (w(D)*w(¬C))) =

((0.9*0.3) + (0.7*0.1)) * ((0.5*0.7) + (0.3*0.5)) + ((0.9*0.7) + (0.3*0.1)) * ((0.5*0.3) + (0.7*0.5)) =

(0.27 + 0.07) * (0.35 + 0.15) + (0.63 + 0.03) * (0.15 + 0.35) =
(0.34) * (0.50) + (0.66) * (0.50) = 0.17 + 0.33 = **0.50**